MIL-L-28948A(MC)
1 September 1982
SUPERSEDING
MIL-L-28948(MC)
7 May 1970

MILITARY SPECIFICATION

LANTERN, ELECTRIC: FLUORESCENT, HAND

This specification is approved for use by U. S. Marine Corps, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a fluorescent, electric, hand lantern using a 110-volt (V) alternating current (ac) source, or a direct current (dc) dry battery source.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-396	- Plastic Mounting and Extrusion Material, Polystyrene.
PPP-B-566	- Box, Folding, Paperboard.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.
PPP-T-76	- Tape, Packaging, Paper (for Carton Sealing).

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document should be addressed to Commandant of the Marine Corps, Headquarters Marine Corps (Code LMA-1), Washington, DC 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document or by letter.

FSC 6230

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MIL-B-18	- Batteries, Dry.
MIL-P-116	- Preservation, Methods of.
MIL-L-970/10	- Lampholder, Miniature Bi-pin for Fluorescent Lamps, 250 Volt, 75 Watt.
MIL-M-13231	- Marking of Electronic Items.
MIL-F-14072	- Finish for Ground Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS15588	- Lamps, Fluorescent, T-5, Miniature Bi-pin.

(Copies of specifications and standards required by suppliers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish five sample units for first article inspection and approval. The first article shall be a prototype sample or it may be a standard production item from the contractor's current inventory.

3.2 Material. The lantern shall be made from materials specified herein. Lantern may contain reclaimed materials provided such materials will not jeopardize its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use in the item.

3.2.1 Finish. All metal parts of the lantern shall be corrosion resisting or have a corrosive-resistant finish applied by plating or enameling.

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3.2.2 Dissimilar metals. The use of dissimilar metals in intimate contact with each other shall be avoided wherever possible. When the use of dissimilar metals cannot be avoided, an interposing material which will eliminate or minimize galvanic corrosion shall be used. Dissimilar metals shall be as defined in MIL-STD-889. All metal surfaces shall be treated and finished in accordance with MIL-F-14072.

3.2.3 Case material. The case for the lantern shall be molded from high-impact polystyrene conforming to L-P-396, type I or IV, class I.

3.2.4 Lamp protector. The lamp protector shall be made of clear acrylic, commercial grade, suitable for the intended use and free from detrimental defects. Such defects include scratches, chips, cracks, striae, or crazing.

3.3 Design and construction. The lantern shall consist of a plastic case, fluorescent lamp, lamp protector, reflector and batteries. An external power cord for operation of the lantern from 110-V ac source shall be included with each lantern. Each unit shall be provided with an ON-OFF switch, a high-low intensity switch, a battery-ac operation switch, and a 110-V ac input. The switching arrangement of the dc dry cell battery source shall be isolated from the 110-V ac source at all times.

3.3.1 Case. The lantern case shall be of two-piece molded construction, reinforced and of sufficient thickness and strength to adequately contain the batteries and other components without bulging, opening at the interface, cracking or otherwise becoming disfigured. The case shall be designed with a carrying handle integrally molded into the case during construction.

3.3.2 Reflector. The reflector shall be metal of suitable weight and composition to hold the lamp protector firmly in place. The minimum thickness shall be 0.020 inch (0.508 mm). Two reinforcing plates shall be attached by rivets, or other suitable means, to the outside of the reflector. The inside of the reflector shall be suitably plated and polished to a high reflecting, nontarnishing surface.

3.3.3 Electrical safety. Metal parts such as receptacles, rivets, screws, reflector, panels, and other exposed components shall be constructed to prevent presence of any voltage on the exterior surface of the lantern.

3.3.4 Battery operation. The lantern shall operate on two dry batteries of 67.5 V each, minimum nominal performance shall be in accordance with MIL-B-18. The lantern shall be tested for operation on battery power as specified in 4.10.

3.3.5 Receptacle access. The receptacle shall be accessible without the removal of any panels or covers. A 110-V ac receptacle shall be provided for the attachment of the external power cord.

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3.3.6 External power cord. A power cord for use when operating the lantern from external 110-V ac source shall be provided with each lantern.

3.3.7 Lampholder. The lampholder shall be capable of holding and retaining the lamp in proper position. The lampholder shall be miniature bi-pin type in accordance with MIL-L-970/10.

3.3.8 Lamp. The lamp shall be fluorescent, miniature bi-pin type, 8 watts, in accordance with MS15588.

3.4 Performance. The lantern shall operate from either dry battery or 110-V ac power to provide a source of general lighting. For battery operation, a current-limiting device and associated switch shall be provided to permit operation of the lantern at approximately 20 percent reduced intensity for conservation of power.

3.5 Dimensions. The maximum dimensions of the lantern shall be as shown in Figure 1. Thickness of the lamp protector shall be 0.065 ± 0.005 inch (1.65 ± 0.127mm).

3.6 Marking. The lantern shall be marked for identification in accordance with MIL-STD-130. External functional components shall be marked to permit identification of the desired function in accordance with MIL-M-13231.

3.7 Workmanship. Workmanship of the lantern shall be of a quality prevailing among manufacturers normally producing lanterns of the type specified herein. All wearing surfaces shall be of materials which will insure a minimum of friction and wear. All parts of the internal mechanism shall be readily accessible for repair and replacement.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspections specified herein are classified as follows:

1. Materials inspection (see 4.4).
2. First article inspection (see 4.5).
3. Quality conformance inspections (see 4.7).
4. Inspection of packaging (see 4.9).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in "Test Conditions" of MIL-STD-810.

4.4 Materials inspection. Materials inspection shall consist of certification, supported by verifying data that the materials specified for use are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.5 First article inspection. When first article is required, inspection shall be performed by the contractor after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.5.1 First article sample. Statistical sampling and inspection of the first article shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be 1.5 percent defective for major defects and 7.5 percent defective for minor defects.

4.5.2 First article inspection routine. The first article sample(s) shall be subjected to the inspection routine as specified in Table I and Table II.

4.5.3 Examination or test. Allowable failures shall be as specified in Table I. One or more major defects shall be cause for refusal to grant first article approval.

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TABLE I - Defects.

Examine	Defect(s)	Classification Major	Minor
Finish	Not manufacturer's standard finish		X
Design, construction, and workmanship, general (applicable) to all components and assemblies)	Design not type specified.	X	
	Assembly or fit of parts or components improper or not as specified herein.	X	
Switches	Switches too loose, too tight, or not functioning properly.	X	
Lamp Holder	Missing or not as specified.	X	
Marking	Not as specified.		X
Power Cord	Plug connectors not as specified.		X
Reflector	Not plated or polished.		X
Dimensions	Not as specified herein.		X

4.6 Inspection data. Unless otherwise specified (see 6.2), inspection data shall be recorded and made available to the acquisition activity for 12 months after the inspection.

4.7 Quality conformance inspection

4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections as specified in Table II.

4.7.1.1 Inspection lot. An inspection lot shall consist of items produced under essentially the same conditions, and offered for inspection at the same time.

4.7.1.2 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot, rework it to correct the defects, or screen out the defective units as applicable, and reinspect. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots. Rejected lots shall be reinspected, using tightened inspection requirements of MIL-STD-105. A lot shall not be resubmitted more than one time.

4.7.2 Group A and B sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for normal inspection, single sampling. The inspection level shall be II and the AQL shall be 1.5.

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TABLE II - Quality conformance inspections.

Inspection	Requirement paragraph	Test method paragraph
Group A		
Visual examination- - - - -	3.3, 3.6	4.5.3
Design, construction, and workmanship - - - - -	3.3	4.5.3
Group B		
Lamp Protector - - - - - -	3.5	4.8.1
Electrical Safety - - - - -	3.3.3	4.8.2
Battery Starting- - - - - -	3.3.4	4.8.3.1
Continuous Battery Operation - - - - - - - -	3.4	4.8.3.2
Intermittent Battery Operation - - - - - - - -	3.4	4.8.3.3
Continuous ac Operation - -	3.3.6	4.8.4.2

4.8 Test methods. Examination and tests shall be performed as specified herein. Other test methods may be substituted upon written approval by the procuring activity, provided the supplier has demonstrated that the results obtained properly correlate with those obtained using the methods specified. The supplier must also demonstrate that such substitution does not relax the requirements of this document.

4.8.1 Lamp protector. Each lamp protector shall be measured for thickness in the following manner:

(a) Two imaginary lines shall be assumed along the length of the lamp protector, positioned a distance in from each edge approximately equal to one-third of the full width of the lamp protector.

(b) Along each line, at approximately equidistant spacing, measurements shall be taken at six points.

(c) Each measurement shall be 0.065 $\pm$ 0.005 inch (1.65 $\pm$ 0.127 mm).

(d) An average of the 12 measurements shall be calculated.

(e) The average measurement shall be 0.065 $\pm$ 0.002 inch (1.65 $\pm$ 0.051 mm).

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4.8.2 Electrical safety. The lantern shall be tested when not operating, and during operation, from both battery and ac source. The test instrument shall be a voltmeter capable of measuring both ac and dc voltage with a minimum impedance of 5,000 ohms per V. The maximum range setting of the voltmeter during the tests shall be 30 V full scale. All external metal surfaces, electrically separated, shall be tested with reference to each other. When being tested during battery operation, the exposed connector pins in the ac receptacle shall be included as a point of test. Painted metal surfaces to be tested shall be scraped or scratched sufficiently to permit electrical contact by the test prods. There shall be no measurable voltage between any two points tested.

4.8.3 Battery operation. The lantern shall be tested on battery power for starting, and intermittent and continuous operation as specified herein. Each battery operation test shall begin with fresh batteries installed in the lantern.

4.8.3.1 Battery starting. The lantern shall be tested for starting operation from battery power source. The cycle of operation shall be 2-1/2 minutes on and 1/2 minute off; variation shall be not greater than $\pm$ 10 seconds. This cycle shall be repeated 40 times for a total test period of 2 hours. Failure of the lantern to light or continue to operate at any time during the test shall be cause for rejection.

4.8.3.2 Continuous battery operation. The lantern shall operate, at reduced intensity, for a period of 30 hours continuous operation. Batteries shall not be replaced during the test. This is not a battery test but a power consumption test of the lantern to verify that the circuitry and other components are not placing an excessive drain on the batteries. Failure of the lantern to operate for the duration of the test shall be cause for rejection.

4.8.3.3 Intermittent battery operation. The lantern shall be tested for intermittent battery operation. The schedule of operation shall be 4 hours on, and 20 hours off over a period of 312 hours, resulting in total operating time of 52 hours. The lantern shall be operated at low intensity. Failure of the lantern to light at the beginning of any cycle, or failure to operate for the duration of the test shall be cause for rejection.

4.8.4 Alternating current operation. The lantern shall be tested on ac power for starting and continuous operation as specified herein. Each ac operation test shall use the power cord supplied with the lantern for connection to external ac power source.

4.8.4.1 Alternating current starting. The lantern shall be tested for starting operation from ac power source. The cycle of operation shall be the same as the battery starting cycle specified in 4.8.3.1. Failure of the lantern to light or continue to operate at any time during the test shall be cause for rejection.

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4.8.4.2 Continuous ac operation. The lantern shall operate continuously for 50 hours. The first 25 hours of operation shall be on high intensity and the second 25 hours shall be on reduced intensity. Failure of the lantern to operate for the duration of the test, or any evidence of overheating or other circuitry defects or inadequacies, shall be cause for rejection.

4.9 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements specified in Section 5. The sample plan shall be in accordance with MIL-STD-105 for normal inspection, single sampling. The inspection level shall be S-4 and the AQL shall be 2.5.

5. PACKAGING

5.1 Preservation-packaging. Preservation shall be level A or commercial, as specified. (See 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. The components and accessories of the lantern shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. The components and accessories of the lantern shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116.

5.1.1.5 Intermediate packaging. Lanterns, packaged as described in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape, and quantities, shall be of minimum tare and cube and shall contain multiples of five unit packages, not to exceed 50 unit packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 units.

5.1.2 Commercial. Commercial packaging shall be in accordance with MIL-STD-1188. This level shall conform to the contractor's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. The packaged lantern shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joint with tape conforming to PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

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5.2.2 Level B. Each lantern, preserved and unit packaged as specified in 5.1.1, shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix of PPP-B-636.

5.2.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Lanterns packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Lanterns packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1, except that the fiberboard caps shall be class domestic.

5.2.4.3 Commercial. Lanterns packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract, each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

6. NOTES

6.1 Intended use. The lanterns are intended to provide general illumination of a limited area. The lantern is not of waterproof construction; therefore, operation of the lantern on 110-V ac power is not recommended when the operator may be in contact with water, moisture, or excessive dampness. The lantern may be used at temperatures to 0°F (-17°C); however, normal battery life would be greatly reduced at low temperature operation.

6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

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(a) Title, number and date of this specification.
(b) When first article approval is required (see 3.1).
(c) When requirements for inspection data shall be other than as specified in 4.6.
(d) Level of preservation, packaging and packing required (see Section 5).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements Lists (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data Requirements	Applicable DID no.	Options
4.5	First Article Inspection Procedure	DI-T-4901	-------
4.5	First Article Inspection Report	DI-T-4902	-------
4.7	Production/Acceptance Inspection Procedures	DI-T-4903	-------
4.7	Production Inspection Reports	DI-T-4904	

(Copies of DIDs required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. When a first article is required for inspection and approval (see 3.1 and 4.5), it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first article shall be a prototype sample or it may be a standard production item from the contractor's current inventory as specified in 3.1. The first article should consist of 5 units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.5 Changes from the previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing Activity:
Navy - MC

Project No. 6230-N262

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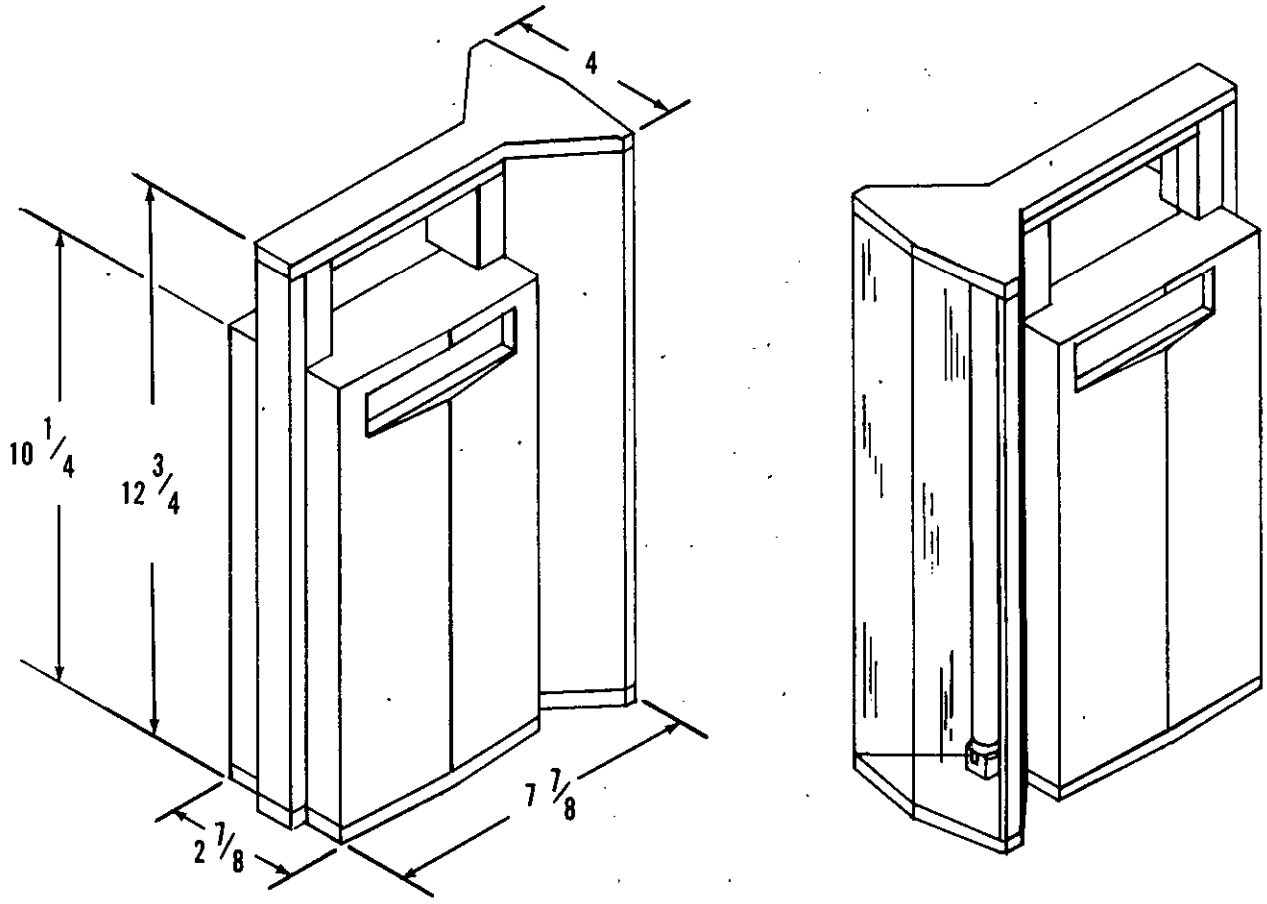


Figure 1. Lantern, Electric, Fluorescent, Hand

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-L-28948A(MC) Lantern, Electric: Fluorescent, Hand

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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